

AWS Retail ETL Pipeline

A serverless ETL pipeline built on AWS to automate the ingestion, validation, transformation, and loading of retail sales data into Amazon Redshift.

Features

- Automated CSV file validation
- Data transformation using AWS Glue
- Data warehouse loading into Amazon Redshift
- Automatic archival of processed files
- Execution logging and error handling

AWS Services

- Amazon S3
- AWS Lambda
- AWS Glue
- Amazon Redshift
- Amazon CloudWatch
- AWS IAM

Repository Structure

```
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├── README.md
├── report.docx
├── file_validation_lambda.py
├── ETL_orchestrator_lambda.py
├── archive_lambda.py
├── architecture.png
└── screenshots/
```

Workflow

1. Upload a CSV file to `raw/sales/`.
2. `file_validation_lambda.py` validates the uploaded file.
3. `ETL_orchestrator_lambda.py` starts the ETL process by:
 4. Running AWS Glue jobs
 5. Executing Redshift SQL scripts

6. Triggering the archive Lambda

7. `archive_lambda.py` archives processed data and cleans temporary folders.

Notes

- SQL scripts used for Redshift loading are not included in this repository.
- In production environments, orchestration can be implemented using **AWS Step Functions** or **Apache Airflow (Amazon MWAA)** instead of a Lambda-based orchestrator.